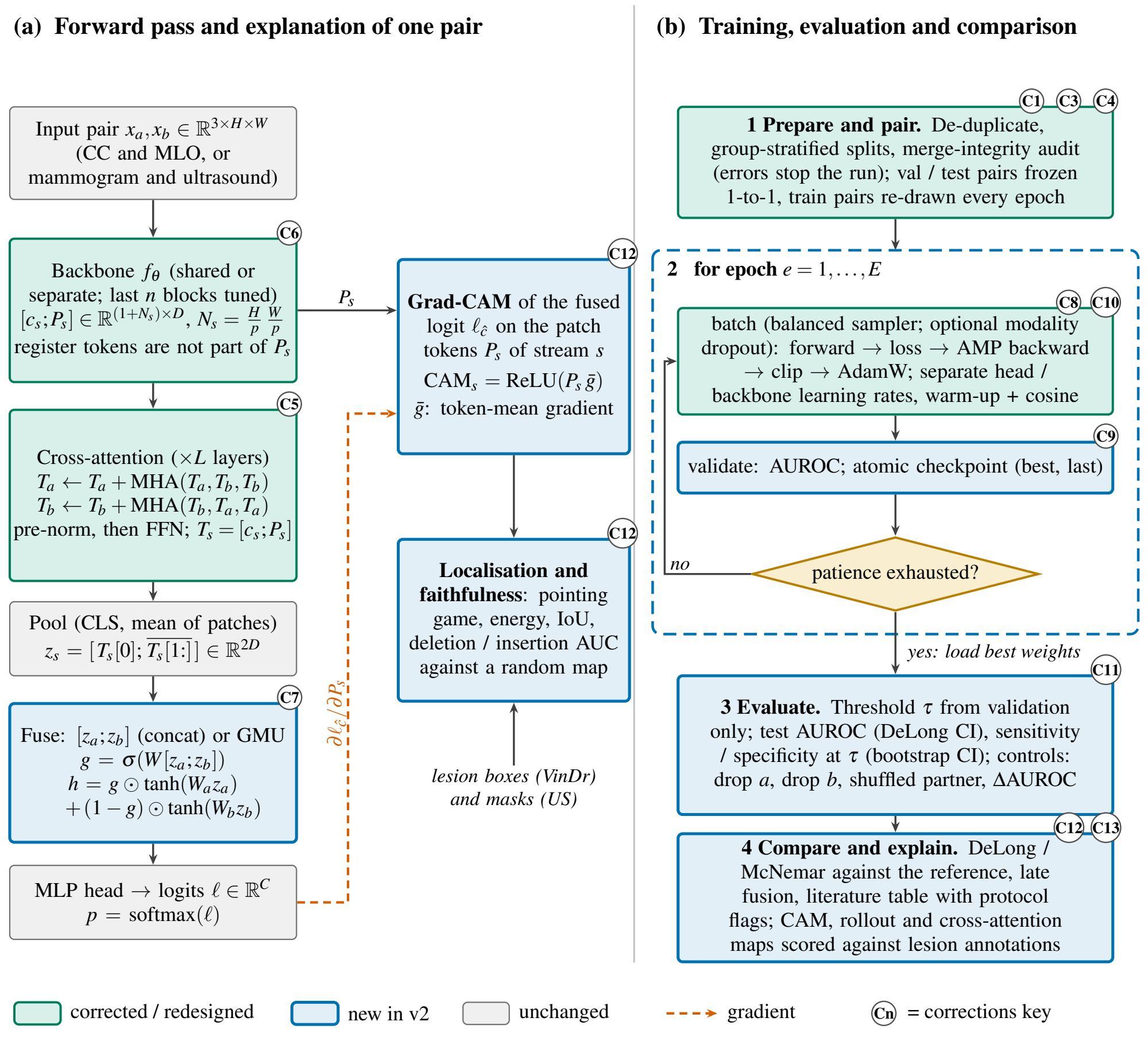

(a) Forward pass and explanation of one pair
(b) Training, evaluation and comparison
Input pair $x_a, x_b \in \mathbb{R}^{3\times H\times W}$ (CC and MLO, or mammogram and ultrasound)
C6
Backbone f_θ (shared or separate; last n blocks tuned) $[c_s; P_s] \in \mathbb{R}^{(1+N_s)\times D}$, $N_s = \frac{H}{p}\frac{W}{p}$ register tokens are not part of P_s
C5
Cross-attention ($\times L$ layers) $T_a \leftarrow T_a + \mathrm{MHA}(T_a, T_b, T_b)$ $T_b \leftarrow T_b + \mathrm{MHA}(T_b, T_a, T_a)$ pre-norm, then FFN; $T_s = [c_s; P_s]$
Pool (CLS, mean of patches) $z_s = [T_s[0]; \overline{T_s[1:]}] \in \mathbb{R}^{2D}$
C7
Fuse: $[z_a; z_b]$ (concat) or GMU $g = \sigma(W[z_a; z_b])$ $h = g \odot \tanh(W_a z_a) + (1-g) \odot \tanh(W_b z_b)$
MLP head $\rightarrow$ logits $\ell \in \mathbb{R}^C$ $p = \mathrm{softmax}(\ell)$
P_s
C12
Grad-CAM of the fused logit $\ell_{\hat{c}}$ on the patch tokens P_s of stream s $\mathrm{CAM}_s = \mathrm{ReLU}(P_s\, \bar{g})$ $\bar{g}$: token-mean gradient
C12
Localisation and faithfulness: pointing game, energy, IoU, deletion / insertion AUC against a random map
lesion boxes (VinDr) and masks (US)
$\partial \ell_{\hat{c}} / \partial P_s$
C1 C3 C4
1 Prepare and pair. De-duplicate, group-stratified splits, merge-integrity audit (errors stop the run); val / test pairs frozen 1-to-1, train pairs re-drawn every epoch
2 for epoch $e = 1, \ldots, E$
C8 C10
batch (balanced sampler; optional modality dropout): forward $\rightarrow$ loss $\rightarrow$ AMP backward $\rightarrow$ clip $\rightarrow$ AdamW; separate head / backbone learning rates, warm-up + cosine
C9
validate: AUROC; atomic checkpoint (best, last)
patience exhausted?
no
yes: load best weights
C11
3 Evaluate. Threshold τ from validation only; test AUROC (DeLong CI), sensitivity / specificity at τ (bootstrap CI); controls: drop a, drop b, shuffled partner, ΔAUROC
C12 C13
4 Compare and explain. DeLong / McNemar against the reference, late fusion, literature table with protocol flags; CAM, rollout and cross-attention maps scored against lesion annotations
corrected / redesigned
new in v2
unchanged
gradient
Cn = corrections key